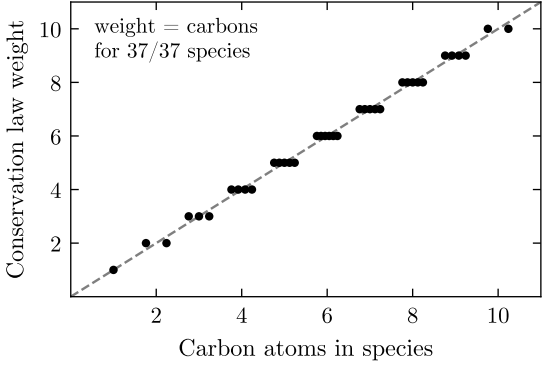
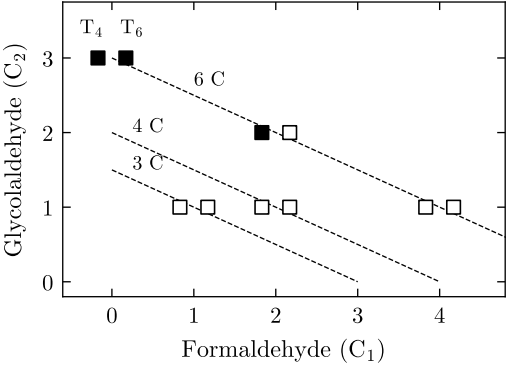
(a)

$2 \times \mathrm{HOCH_2–CHO} \xrightarrow{r_1 \times 2} 2 \times \mathrm{HO–CH{=}CH–OH}$

(b)



(c)